

Module - 1 Revision

1.

Determine the order of the PDE:

$$\frac{\partial^3 u}{\partial x^3} + \frac{\partial^2 u}{\partial y^2} = \sin(x + y)$$

2.

Find the degree of the PDE:

$$\left(\frac{\partial^2 z}{\partial x^2} \right)^4 + \frac{\partial z}{\partial y} = 0$$

3.

Determine the **order** and **degree** of the PDE:

$$\frac{\partial^2 u}{\partial x^2} + \left(\frac{\partial u}{\partial y} \right)^3 = 0$$

4.

In heat transfer analysis, the temperature distribution on a metal plate varies with distance and satisfies

$$\frac{\partial^2 T}{\partial y^2} = x$$

Determine the temperature function $T(x, y)$.

5.

In a railway track system, thermal expansion causes uniform heating along the rail. The temperature distribution satisfies

$$\frac{\partial^2 T}{\partial x^2} = k^2$$

Determine the temperature function $T(x, y)$.

6. Find the complete integral of the PDE: $p - q = 3$

7. Find the complete integral of $z = px + qy + p^2 - q^2$.

8. In image processing, the brightness function $B(x, y)$ of an image satisfies $p^2 + q^2 = x + y$, where p and q are gradient components. **Determine** the brightness function.

9. In computer vision, the brightness function $B(x, y)$ satisfies the relation $q - p + x - y = 0$, where p and q are gradient components. **Determine** the brightness function.

10. Solve $\frac{\partial^2 z}{\partial x^2} - 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = e^{x-y} + e^{2x+4y} + \sin(x + 2y)$

11. Solve $\frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial x \partial y} + 4 \frac{\partial^2 z}{\partial y^2} = e^{2x+y} + \cos(2x + 3y)$

12. Find the general solution of the PDE: $xp + yq = 2z$.

13. Solve $(x - z)p + (y - x)q = z - y$.

14. Solve $(mz - ny)p + (nx - lz)q = (ly - mx)$.

15. Solve $x(z^2 - y^2)p + y(x^2 - z^2)q = z(y^2 - x^2)$